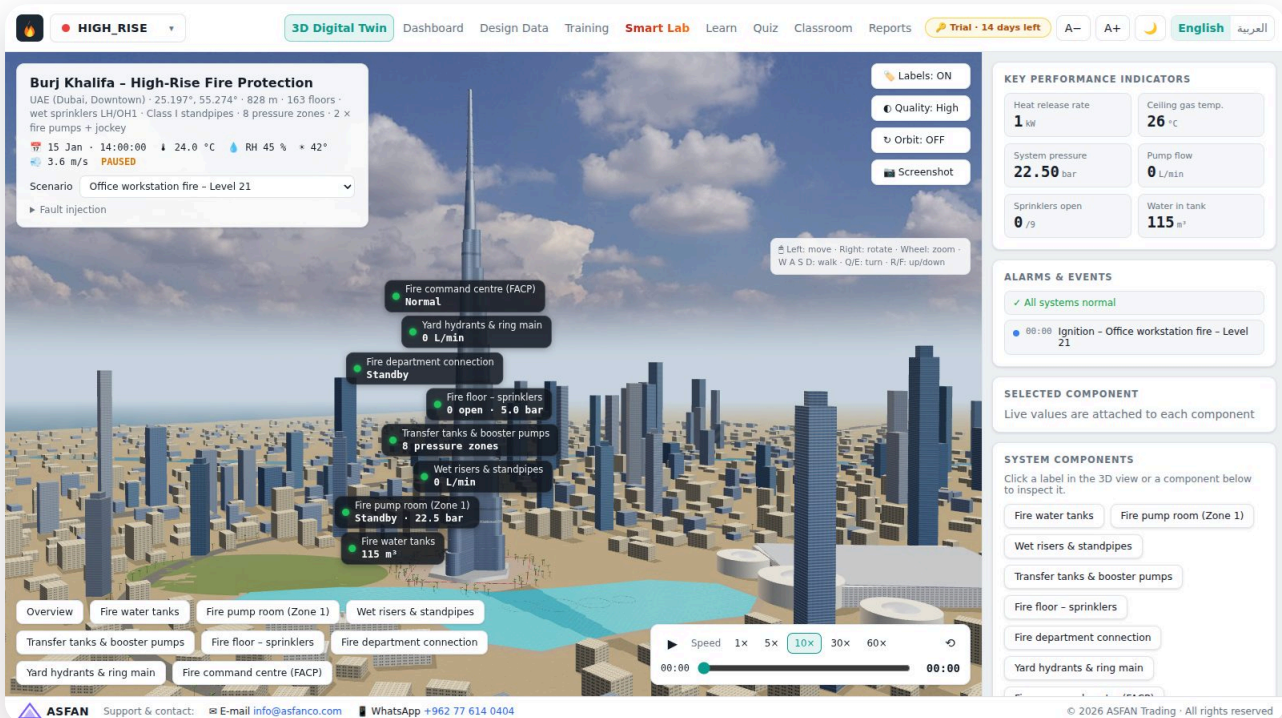




Fire Protection Digital Twin

User Manual · Educational fire-fighting & smart fire-alarm simulator

Version 1.1 · Windows 10 / 11 (64-bit) · English



ASFAN Trading

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Fire Protection Digital Twin is an educational real-time simulator of fire-fighting and fire-alarm systems for universities, training centres and companies. Real sites (Burj Khalifa, Aqaba port, Amman data centre, oil tank farm, airport hangar) are modelled in 3D with their fire-protection systems designed to NFPA standards. A physics engine simulates fire growth, detection, sprinkler operation, pumps, water supply, clean agents and foam — and every value is shown live on the 3D model.

What you can do

- Watch complete fire scenarios in 3D, with live pressures, flows, temperatures and alarms on every component.
- Inject faults (power failure, closed valve, jockey pump failure, leaks...) and diagnose them.
- Edit the design data and run the simulation on your own design, with NFPA compliance checks.
- Practise real procedures in ten interactive 3D training scenes, scored step by step.
- Learn smart (addressable) fire-alarm systems in the Smart Systems Lab: panel, cause & effect, BMS integration, commissioning, maintenance and incident command.
- Study lessons, take quizzes, run classroom challenges, print reports and certificates.

Educational use

Building data are public information; the fire systems are NFPA-based reference designs for teaching. Do not use results for real projects without verification by a qualified engineer.

2 System requirements

	Minimum	Recommended
Operating system	Windows 10 64-bit	Windows 11 64-bit
Processor	Intel Core i5 (8th gen) / Ryzen 5	Core i7 / Ryzen 7
Memory	8 GB RAM	16 GB RAM
Graphics	Integrated GPU with DirectX 11 / WebGL 2	Dedicated GPU (NVIDIA / AMD), 2 GB
Screen	1366 × 768	1920 × 1080 or larger
Disk space	about 500 MB	
Internet	Not required (only for the optional YouTube source-video links)	

3 Installing the program

You receive one of two files from ASFAN:

File	Use
FireProtectionDigitalTwin-Setup-1.1.0.exe	Installer (recommended). Installs into Program Files, creates desktop and Start-menu shortcuts and an uninstaller.
FireProtectionDigitalTwin-Portable-1.1.0.exe	Portable. Runs directly without installation — e.g. from a USB stick or a lab PC without admin rights.

Installation steps

- 1 Double-click `FireProtectionDigitalTwin-Setup-1.1.0.exe`.
- 2 Windows may show a blue “**Windows protected your PC**” window (Microsoft SmartScreen). Click **More info**, then **Run anyway**. This appears because the program is new and is not yet signed with a paid code-signing certificate — it is safe to continue.
- 3 Choose the installer language (English / العربية) and accept the license agreement.
- 4 Choose **for all users** or **only for me**, and the installation folder (the default is fine).
- 5 Click **Install**, then **Finish**. The program starts automatically; a shortcut **Fire Protection Digital Twin** is placed on the desktop.

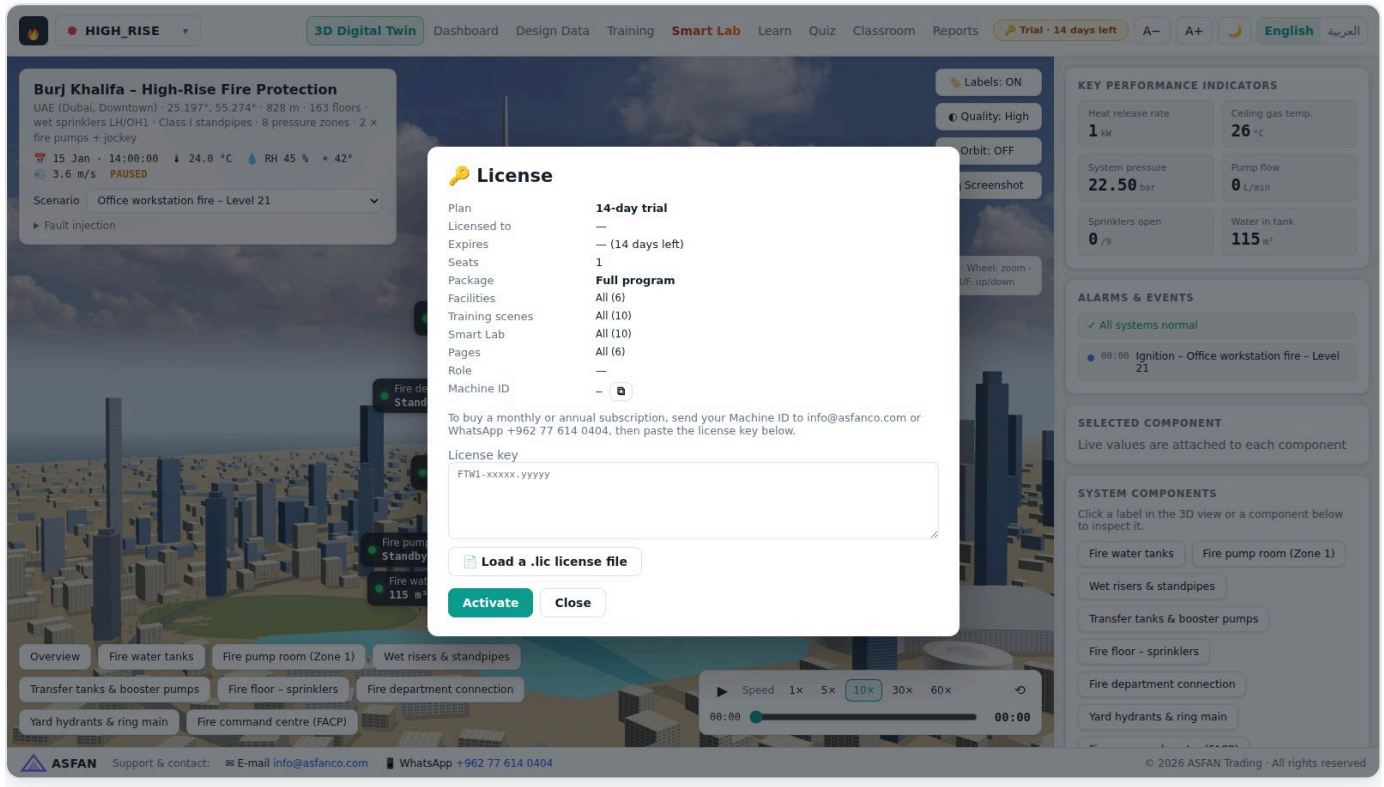
Antivirus

Some antivirus programs may scan the new file for a few seconds or ask for confirmation the first time. Allow the file — it is the official ASFAN installer. If a school network blocks unknown programs, ask the IT administrator to allow the installer or use the portable version.

Updating and uninstalling

- **Update notice:** when the computer is online, the program checks for a new version at start-up (and from **Help** → **Check for updates**). A green bar appears at the top: **A new version is available** — click **What's new** or **Download**.
- **Update:** run the new Setup file — it replaces the old version. Your license, settings and results are kept.
- **Uninstall:** Windows Settings → Apps → Fire Protection Digital Twin → Uninstall.

On the first start the program runs as a **14-day full trial**. The license badge at the top (🔑) shows the remaining days.



The license window: plan, customer, expiry, package contents and the Machine ID

Buying and activating a subscription

- 1 Click the 🔑 badge at the top of the window (or **Help** → **License...**).
- 2 Copy your **Machine ID** with the 📄 button and send it with your order to ASFAN (e-mail or WhatsApp) — only needed if your license should be bound to this computer.
- 3 ASFAN sends you a license file `.lic` (or a key starting with `FTW1-`).
- 4 In the license window click **Load a .lic license file** (or paste the key) and press **Activate**. The badge changes to **Licensed · Monthly / Annual**.

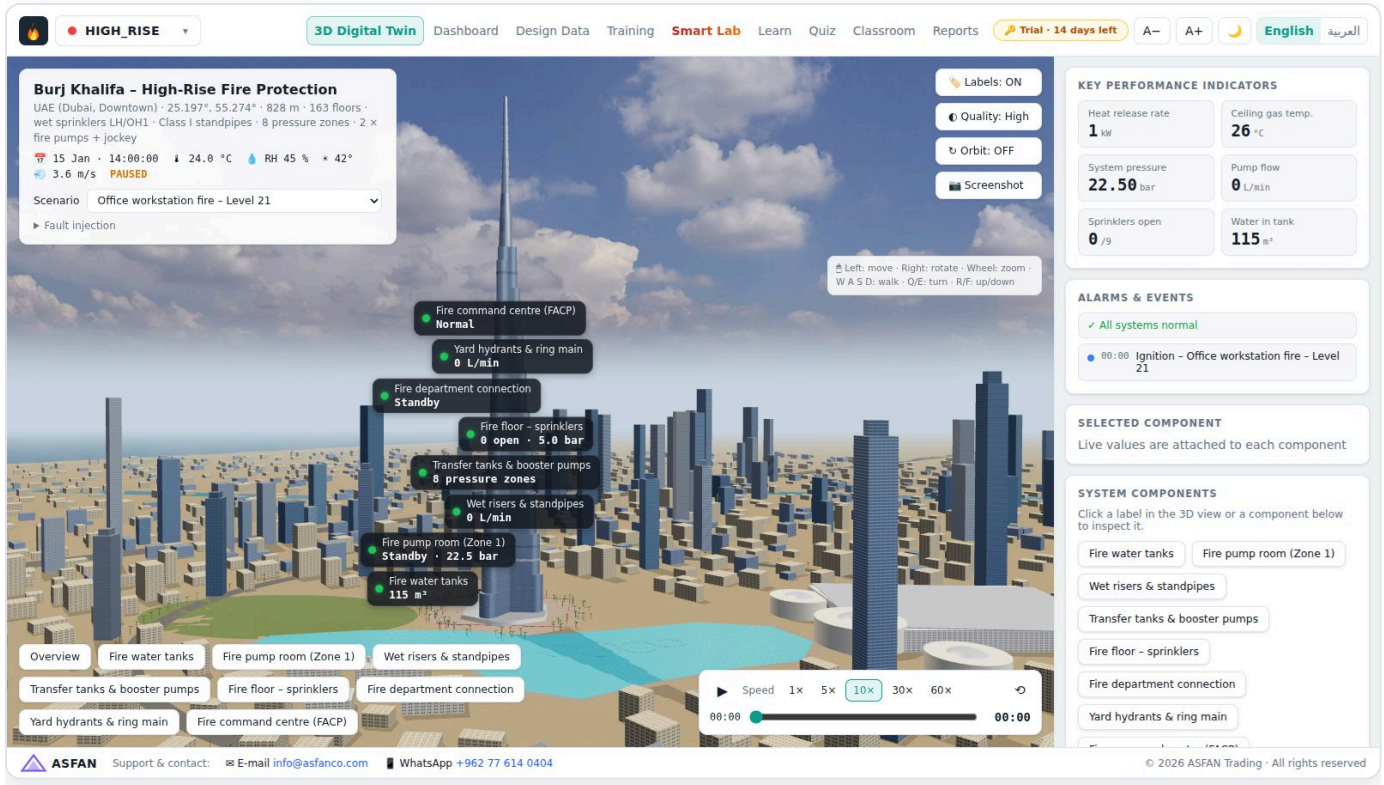
Packages

A license can include the **full program** or only a **package**, for example *Sprinklers & pumps*, *Smart fire alarm*, *Special hazards* or *Basic awareness*, or a custom selection. The license window lists exactly which facilities, training scenes, Smart Lab modules and pages are included. Everything else is shown with a 🔒 lock; clicking it opens the license window so you can upgrade.

Situation	What happens
Subscription expires	The program asks for a new key. Your data and results are kept.
New computer	Unbound licenses can be activated on the new PC; bound licenses need a new key for the new Machine ID.
Supervisor (teacher) license	Sees, exports and clears the results of all students and the instructor dashboard.

Do not change the computer date

Setting the clock back is detected and blocks the program until the date is corrected.



Main screen: top bar, 3D view with live labels, information card, KPIs, alarms and components

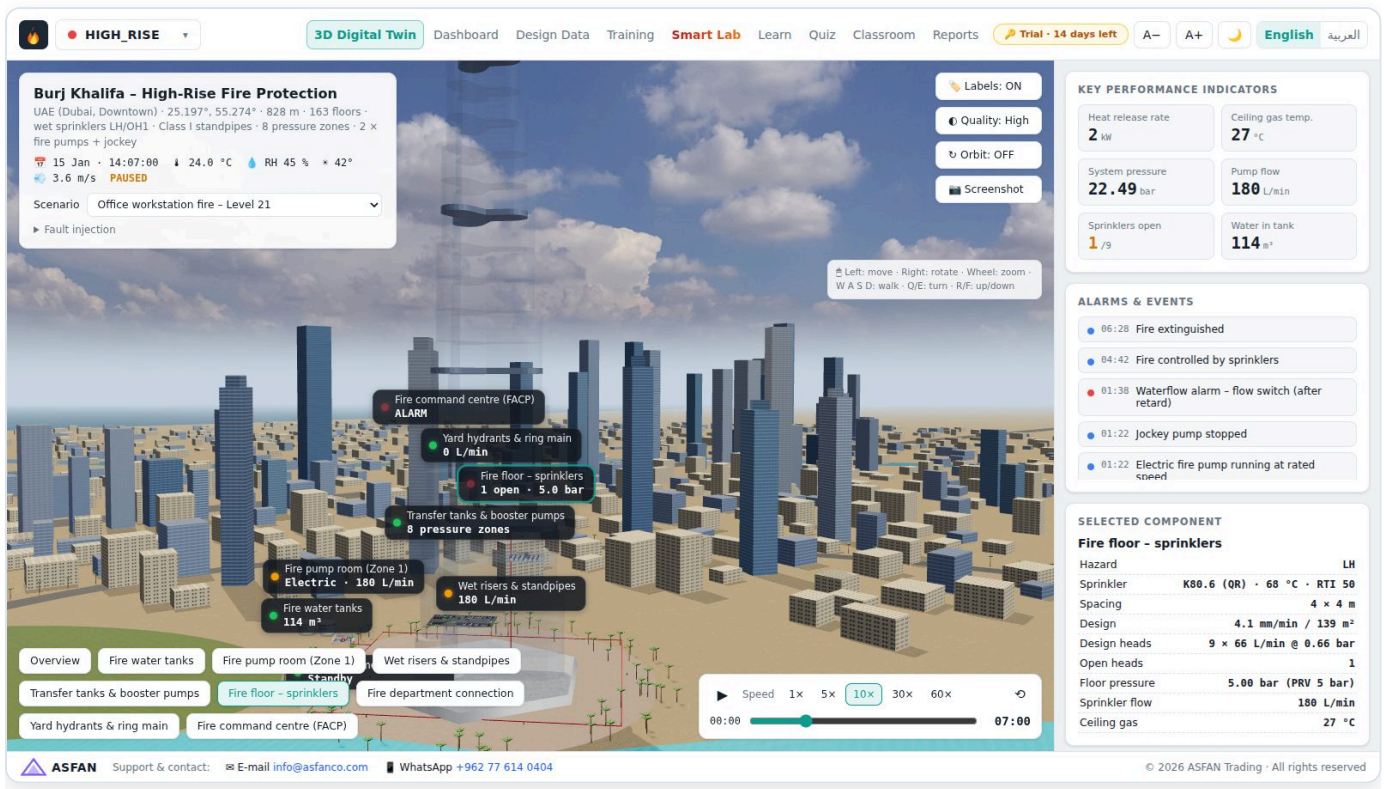
Top bar

Element	Function
Facility selector	Choose one of the reference facilities or Create facility manually (your own building).
Tabs	3D Digital Twin · Dashboard · Design Data · Training · Smart Lab · Learn · Quiz · Classroom · Reports.
🔑 badge	License status and activation.
A- / A+	Smaller / larger text.
🌙 / ☀️	Dark / light theme.
?	Opens this user manual.
English / العربية	Switch the whole program language (Arabic is right-to-left).

Moving in 3D

Action	Mouse / keyboard
Move (pan)	Left button drag
Rotate	Right button drag
Zoom	Mouse wheel
Walk	W A S D
Turn	Q / E
Up / down	R / F
Play / pause simulation	Space

Buttons on the right of the 3D view: **Labels** on/off, **Quality** high/low (use *low* on slow laptops), **Orbit** auto-rotation, and **Screenshot** (also **Ctrl** + **Shift** + **S**).

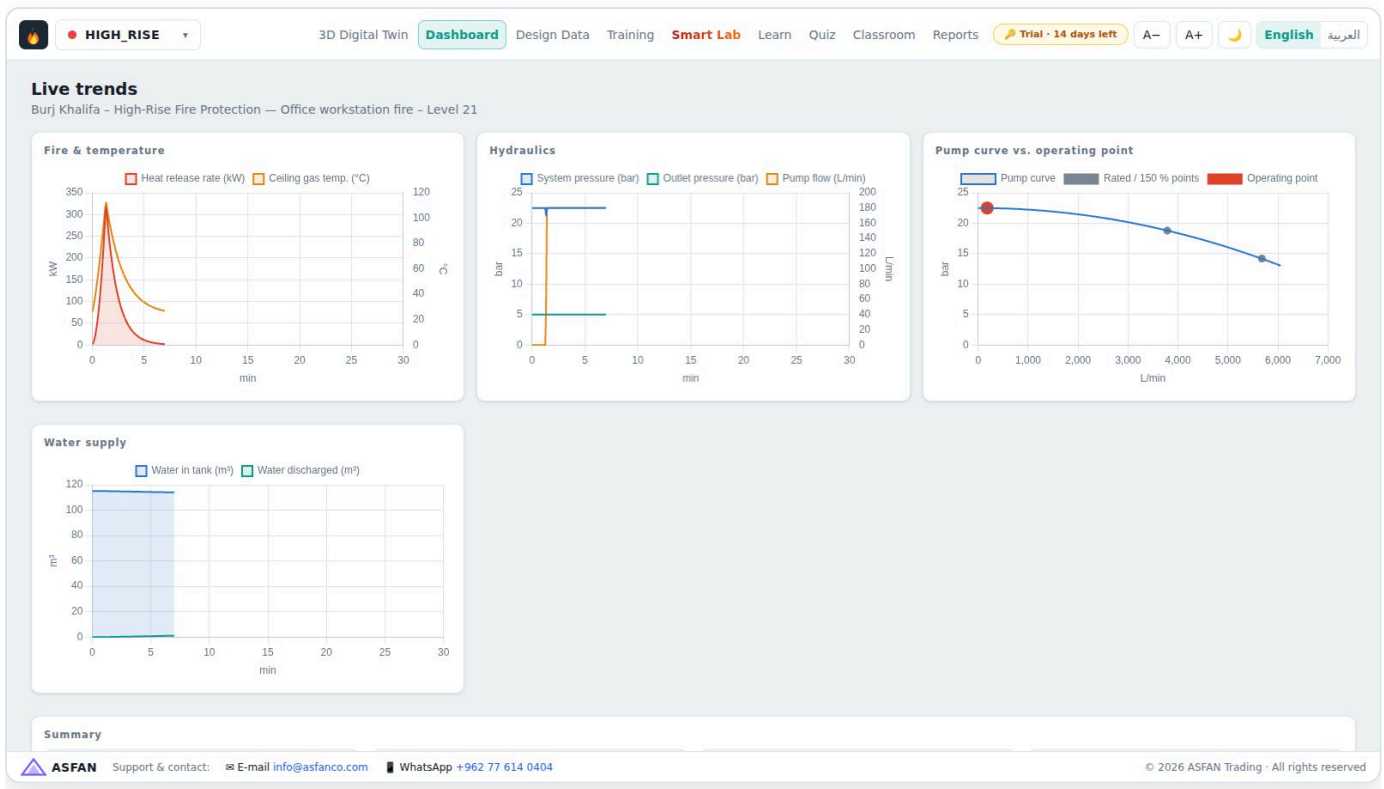


A fire scenario running: sprinklers operating, live pressures and alarms

- 1 Select a **facility** and a **scenario** (e.g. fire on a given floor).
- 2 Press ► to play. Use the speed buttons (1×–60×) and drag the time slider to jump.
- 3 Click any label or component to fly to it; its live values appear in **Selected component**.
- 4 Watch **KPIs** (heat release, gas temperature, pressure, flow, sprinklers open...) and the **Alarms & events** log.
- 5 Open **Fault injection** to add failures (power failure, closed valve, leak, pump failure...) and see how the system reacts.

Create facility manually

Choose it from the facility selector to build your own building (city, occupancy, floors, dimensions, sprinkler K-factor, temperature rating, response type). The design and simulation are generated automatically.



Dashboard: fire, hydraulics, pump curve with moving operating point and water supply

Live trend charts for the running scenario: heat release and gas temperature, system pressure and flow, the fire-pump curve with the moving operating point, and tank/agent/foam supply. Charts follow the simulation time.

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Dashboard
Design Data
Training
Smart Lab
Learn
Quiz
Classroom
Reports
Trial - 14 days left
A-
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English
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Design basis & calculations

Burj Khalifa - High-Rise Fire Protection — Office workstation fire - Level 21

Edit design data - your own values
Change any value, then run the simulation on your data. Changed fields are highlighted; values are saved per facility and scenario.

Sprinkler system (NFPA 13)

Occupancy hazard LH	Sprinkler K-factor (L/min/vbar) 80.6	Sprinkler spacing (square) (m) 4	Response time index RTI ((m·s) ^{1/2}) 50	Temperature rating (°C) 68	Hose allowance (L/min) 379
------------------------	---	-------------------------------------	---	-------------------------------	-------------------------------

Water supply duration (min)
30

Piping & building

Main pipe size (mm) 150	Main pipe length (m) 140	Fire floor elevation above pump (m) 88	Floor PRV setting (0 = none) (bar) 5	Ceiling height (m) 3	Room length (m) 24
Room width (m) 16	Fire position x (m) 13	Fire position y (m) 7			

Fire scenario

Fire growth rate (t²) fast	Peak heat release rate (kW) 3000	Ambient temperature (°C) 24
-------------------------------	-------------------------------------	--------------------------------

Pumps & water

Fire pump rating (0 = auto) (gpm) auto	Pump rated pressure (0 = auto) (bar) 0	Number of duty pumps (0 = auto) 0	Water storage (0 = auto) (m³) 0
---	---	--------------------------------------	------------------------------------

▶ Run simulation with my data
✓ Recalculate
↶ Reset to reference design
↓ Export JSON
↑ Import JSON

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Design Data: NFPA calculation steps, editable values and compliance checks

Shows every design calculation step (NFPA 13 density/area, K-factor, Hazen–Williams, pump selection, NFPA 2001 agent quantity, NFPA 11 foam...) in SI and US units. Change any value in the editor, press **Apply** or **Apply & run**, and the simulation runs on your design. The compliance list shows which NFPA rules pass or fail. **Reset** returns to the reference design.

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Training

Interactive 3D training scenes based on the course videos: explore every component, then perform the procedure step by step on live physics. Results are saved to the Classroom.

New: Smart Systems Lab
Addressable panel, cause & effect, BMS integration, commissioning, predictive maintenance, incident command and certificates.

Open the Smart Lab →

Fire pump room walk-through & start sequence
NFPA 20 pump room: tank, suction, jockey / electric / diesel pumps, controllers, relief valve, test loop. Operate the demand and watch the pressure-switch sequence.

▶ Start ▶ Source video

Annual fire pump flow test (NFPA 25)
Run the pump at churn, 100 % and 150 % of rated flow through the flow-meter loop, record net pressure and compare with the acceptance curve.

▶ Start ▶ Source video

Dry-pipe valve: trip test & reset
Air holds the clapper closed; opening the inspector's test lets air escape, the valve trips and water must reach the test outlet within the NFPA 13 limit. Then restore the system step by step.

▶ Start ▶ Source video

Floor control valve assembly: alarm & tamper tests
Zone control valve with tamper switch, waterflow switch with retard, gauge, test-and-drain with sight glass and PRV. Run the waterflow alarm test and the valve supervisory test.

▶ Start ▶ Source video

Portable extinguishers: placement & PASS
Place 2-A:10-B:C extinguishers on a light-hazard office floor so every point is within 22.9 m (75 ft) travel distance (NFPA 10), then fight a fire using P-A-S-S.

▶ Start ▶ Source video

Hose drill: landing valve & fire hose cabinet
Two-person team: open the cabinet, run out the 65 mm hose without kinks, couple and tug-test, fit the branch nozzle, open the landing valve slowly and attack the fire at its base.

▶ Start ▶ Source video

Sprinkler types & response race
Compare pendent, upright, sidewall, concealed, recessed, ESFR and open heads, then light a t³ fire under six bulbs of different temperature rating and RTI and watch which operates first.

▶ Start ▶ Source video

Stairwell pressurization (NFPA 92)
Keep smoke out of the escape stair: set the supply fan so $\Delta P \geq 12.5$ Pa, keep door-opening force ≤ 133 N, and see what happens when doors are opened. $Q = 0.839 A \sqrt{\Delta P}$.

▶ Start ▶ Source video

FM-200 room: design calculator & discharge sequence
Size the agent live ($W = V/S \times C/(100-C)$) with NOAEL/LOAEL warnings, then run the release sequence: cross-zoned detection, 30 s pre-discharge, HVAC shutdown, 10 s discharge and hold.

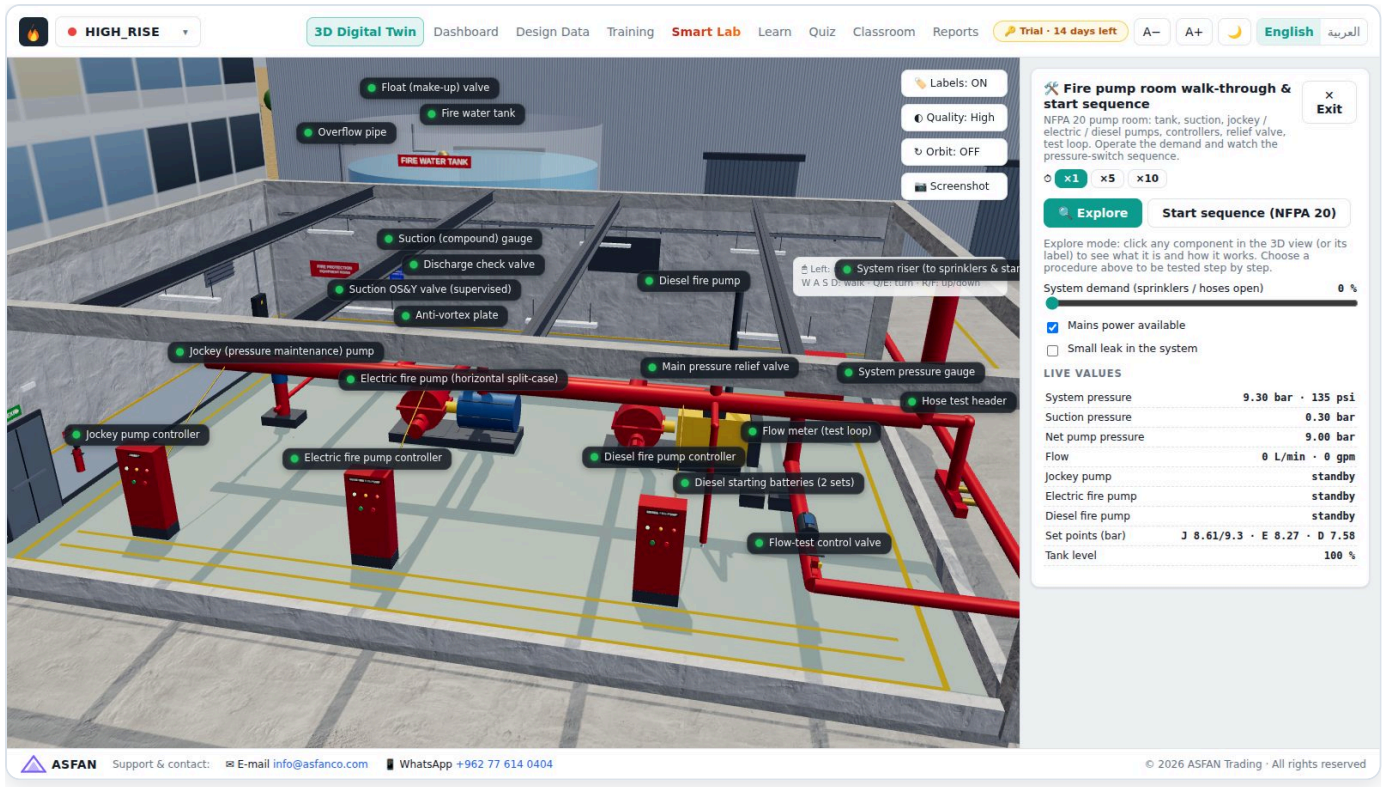
Installation mode: detectors & sprinklers
Lay out smoke detectors ($0.7 \times S$ rule) and light-hazard pendent sprinklers on a real office ceiling with a deep beam, supply diffusers, an exposed duct and a column — live coverage, NFPA 72 / NFPA 13 obstruction rules, and an installer's layout to correct.

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The Training page with the ten interactive 3D scenes

Scene	What you practise
Fire pump room	NFPA 20 components and the jockey → electric → diesel start sequence, manual stop.
Annual pump flow test	Churn, 100 % and 150 % flow, acceptance curve (NFPA 25).
Dry-pipe valve	Full trip test with water-delivery time and the reset procedure.
Floor control valve	Waterflow alarm test (retard) and tamper / supervisory test.
Portable extinguishers	Placement to NFPA 10 travel distance and the P-A-S-S technique.
Hose drill	Landing valve, couplings and two-person team operation.
Sprinkler types	Pendent, upright, sidewall, concealed, ESFR... and a response race.
Stair pressurization	NFPA 92 pressure difference, door-opening force, relief damper.
FM-200 room	NFPA 2001 calculator and the cross-zoned discharge sequence.
3D installation mode	Placing detectors and sprinklers on a real ceiling with obstruction rules.



A training scene (fire pump room) with the procedure panel on the right

How a scene works

- 1 **Explore:** click any component (or its label) to read what it is and how it works.
- 2 Choose a **procedure**. The steps appear on the right; the current step is highlighted.
- 3 Perform each step by clicking components, moving sliders or answering questions. Wrong actions count as mistakes.
- 4 Use ×1 / ×5 / ×10 to speed up waiting times. At the end you get a score; results are saved to the Classroom.

Labels: ON
Quality: High
Orbit: OFF
Screenshot

Fire pump room walk-through & start sequence
 NFPA 20 pump room: tank, suction, jockey / electric / diesel pumps, controllers, relief valve, test loop. Operate the demand and watch the pressure-switch sequence.

Explore Start sequence (NFPA 20)

- Check that the suction supply is open: click the supervised suction OS&Y valve.
- Identify the jockey (pressure-maintenance) pump.
- Open a small demand ($\approx 2\%$). The jockey must start at 8.61 bar.
- Increase the demand to $\approx 60\%$ (sprinklers operating). The electric fire pump starts at 8.27 bar.
- Close the demand to 0% . Does the electric fire pump stop by itself?
- With the demand closed, stop the electric pump: click its controller (STOP).
- Simulate a mains power failure: switch off "Mains power available".
- Open the demand again ($\approx 60\%$). The electric pump cannot start; the diesel cranks at 7.58 bar.
- Close the demand, restore mains power, then stop the diesel pump at its controller.

System demand (sprinklers / hoses open) 0 %

☒ Mains power available
☐ Small leak in the system

LIVE VALUES
 System pressure 9.30 bar · 135 psi

10 Smart Systems Lab (advanced)

The **Smart Lab** tab trains engineers on smart (addressable) fire-alarm and life-safety systems. All ten modules share one live system: a fault planted in the panel appears in the building, the BMS, the monitoring centre and the commissioning records. A ✓ in the left menu marks completed modules.

The screenshot displays the 'Smart Systems Lab' interface. On the left, a sidebar lists ten modules, with the first one, 'Addressable Fire Alarm Panel Lab', marked as completed with a green checkmark. The main area is titled 'Addressable Fire Alarm Panel Lab' and includes a description: 'Operate a real-style addressable FACP connected to a 3-storey building; test every device in the field, cut and short the SLC loop to see Class A and isolators at work, program addresses with auto-learn, and diagnose hidden faults like a service engineer.' Below this, there are tabs for 'Live panel & building', 'Programming', 'Guided exercises', 'Inspector challenge', and 'Devices: install & wiring'. The 'Live panel & building' tab is active, showing a digital representation of the 'ASFAN FX-3000' fire alarm control panel. The panel's LCD displays 'SYSTEM NORMAL', 'Loop 1: 61 devices OK Class A', 'AC ON BATT 27.3V', and 'ACCESS LEVEL 1'. Below the LCD are buttons for 'ACKNOWLEDGE', 'ALARM SILENCE', 'RESOUND', 'SYSTEM RESET', 'DRILL', 'LAMP TEST', and 'BUZZER'. To the right of the panel is a floor plan diagram of a 3-storey building with various rooms and a network of red lines representing the SLC loop. A toolbar above the floor plan allows users to 'Select / test', 'Cut cable', 'Short', 'Repair', and 'Add device'. A legend below the floor plan identifies symbols for SLC loop, NAC circuit, Isolator, Call point, Module, and Detector. At the bottom of the interface, there is a footer with 'ASFAN' logo, contact information, and a copyright notice for 2026 ASFAN Trading.

Module 1 – addressable fire alarm panel with the building plans and SLC loop

1 · Addressable panel lab

A realistic panel (LCD, LEDs, **ACKNOWLEDGE**, **ALARM SILENCE**, **SYSTEM RESET**, **DRILL**, lamp test and the access-level key switch). Click a device on the plan to test it (smoke, heat, call point, waterflow, valve). Use the toolbar to **cut** or **short** the loop, compare Class A and B, see isolators work, add devices and run **auto-learn** in the Programming tab. **Guided exercises** are detected automatically; the **Inspector challenge** hides faults for you to find. Silence and reset need access level 2 — turn the key.

HIGH_RISE

3D Digital Twin

Dashboard

Design Data

Training

Smart Lab

Learn

Quiz

Classroom

Reports

Trial - 14 days left

A-

A+

English

العربية

ADVANCED

Smart Systems Lab

Install, program, integrate, commission and maintain an addressable fire-alarm system.

1

Addressable Fire Alarm Panel Lab

SLC loop, isolators, faults, progr...

2

Cause & Effect Matrix

Program the smart logic & run fl...

3

BMS & Building Integration

BACnet/Modbus, lift, voice evac...

4

Installation Mode (3D)

Detector & sprinkler layout rules

5

Commissioning & Acceptance Testing

Acceptance tests & record of co...

6

Predictive Maintenance

Condition monitoring, ITM & wor...

7

Incident Commander

Decisions under pressure, casca...

8

Modern Detection & Suppression Technologies

ASD, beam, flame, LHD, mist, pr...

9

Engineering Tools

Hydraulics, batteries, voltage dr...

10

Learning Paths & Certificates

Levels, final exam, certificates

✓

Commissioning & Acceptance Testing

Take the installed system through acceptance exactly as a commissioning engineer does: pre-functional inspection and cable tests, a device-by-device functional test on the live panel, audibility and visibility survey, battery and power tests, the sprinkler hydrostatic test — and issue the NFPA 72 Record of Completion.

NFPA 72 §14.4 • T.14.4.3.2

NFPA 72 §7.5.6

NFPA 72 §18.4 • §18.5

NFPA 72 §10.6

NFPA 13 §29.2.1

1 • Pre-functional

2 • Device functional test

3 • Audibility & visibility

4 • Battery & power

5 • Hydrostatic test

6 • Record of Completion

Checklist items

0/14

Flagged entries

0

Installation defects recorded

0

Loop class

A

Programmed devices

61

PRE-FUNCTIONAL INSPECTION & CABLE TESTS

NFPA 72 §14.4 • T.14.4.3.2

Item / acceptance criterion	Measured	Verdict
○ Addressable devices, panel and surge protectors disconnected before insulation test Mfr. / NFPA 72 §14.2.10	☐ Verified on site	
○ SLC loop cable insulation resistance (conductors → earth) ≥ 1 MΩ @ 500 V DC • NFPA 72 T.14.4.3.2 (21)	— MΩ Measure	Pass Fail
○ NAC1 cable insulation resistance ≥ 1 MΩ @ 500 V DC • NFPA 72 T.14.4.3.2 (21)	— MΩ Measure	Pass Fail
○ NAC2 cable insulation resistance ≥ 1 MΩ @ 500 V DC • NFPA 72 T.14.4.3.2 (21)	— MΩ Measure	Pass Fail
○ SLC loop resistance, panel OUT → IN (continuity) ≤ 40 Ω • NFPA 72 T.14.4.3.2 (21)	— Ω Measure	Pass Fail
○ Earth-fault monitor: loop resistance to earth (panel reading) ≤ 50 kΩ • NFPA 72 §12.6	— kΩ Measure	Pass Fail
○ NAC1 end-of-line resistor 4.7 kΩ ± 5 % • NFPA 72 §12.6	— kΩ Measure	Pass Fail
○ NAC2 end-of-line resistor 4.7 kΩ ± 5 % • NFPA 72 §12.6	— kΩ Measure	Pass Fail
○ Loop scan: devices answering = devices programmed, no double address = programmed • NFPA 72 §14.4.1	— dev. Measure	Pass Fail
○ Max. smoke-detector spacing, smooth ceiling (corridor) ≤ 18 m • NFPA 72 §15.2.5.5	— m Measure	Pass Fail

TEST INSTRUMENT

Why disconnect first?

A 500 V DC insulation test destroys the electronics of addressable detectors, modules and the panel. Megger the cables with devices removed from their bases and conductors lifted at the panel.

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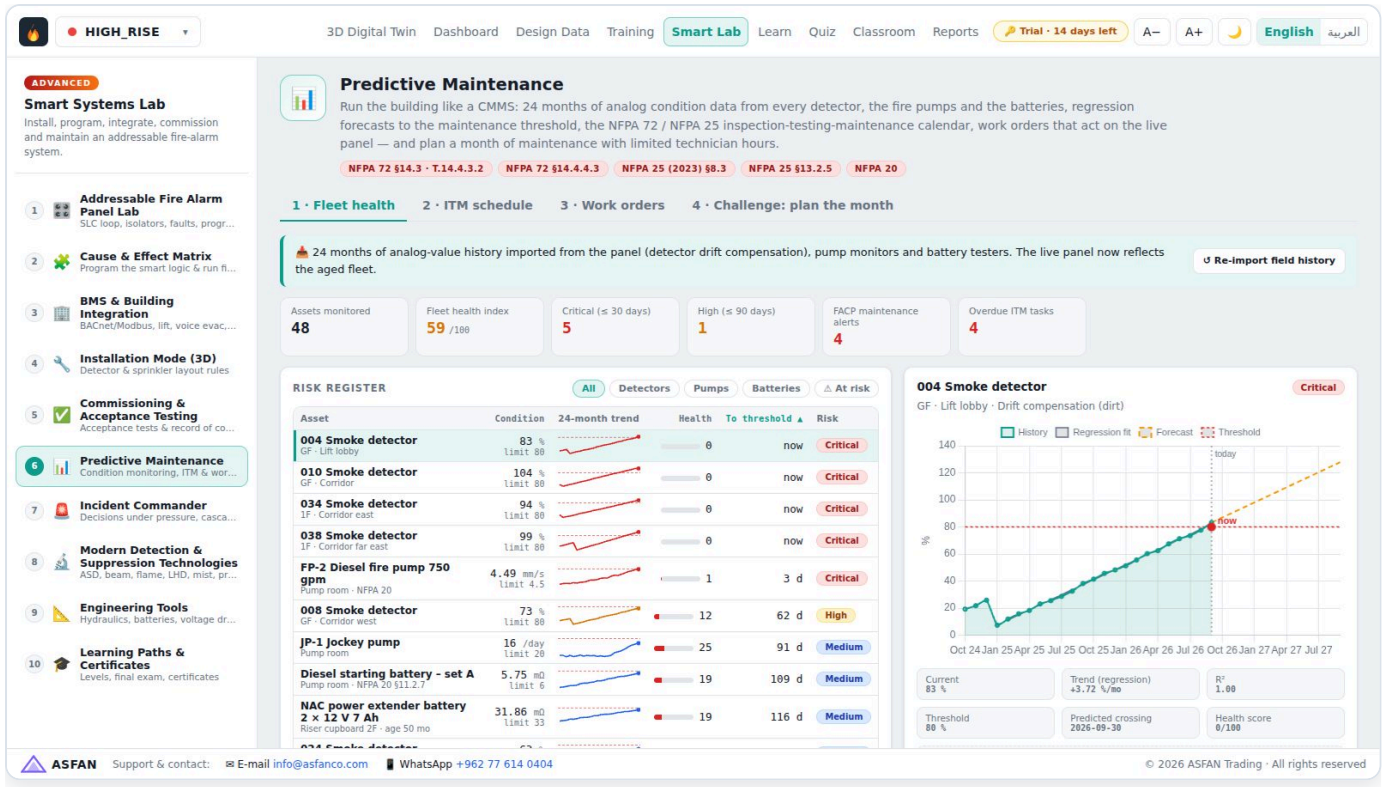
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Module 5 – commissioning: device tests, audibility, batteries, hydrostatic test, record

6 • Predictive maintenance

Condition data for detectors, pumps and batteries with trend forecasts, the NFPA 72 / NFPA 25 inspection-testing-maintenance calendar, work orders that really clean detectors on the panel, and a monthly planning challenge.



Module 6 – fleet health, risk register and forecast

7 · Incident commander

Timed decisions in fires with cascading failures (night server-room fire, office-hours kitchen fire). Afterwards **Replay & debrief** shows your timeline, the best actions and their code basis.

8 · Modern detection & suppression

Simulators for aspirating smoke detection (VESDA), beam detectors, UV/IR flame detectors, linear heat cable, water mist and pre-action systems (non-, single- and double-interlock), plus a technology-selection challenge.

10 · Learning paths & certificates

Enter your name, follow a path (Technician, Engineer, Supervisor, Designer), complete the required modules, pass the final exam ($\geq 80\%$) and print a certificate with a verification code. Supervisors see the instructor dashboard with each trainee's progress.

**HIGH_RISE**

3D Digital TwinDashboardDesign DataTrainingSmart LabLearnQuizClassroomReportsTrial - 14 days leftA- A+Englishالعربية

ADVANCED

Smart Systems Lab
Install, program, integrate, commission and maintain an addressable fire-alarm system.

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ASD, beam, flame, LHD, mist, pr...

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Engineering Tools
Hydraulics, batteries, voltage dr...

10

Learning Paths & Certificates
Levels, final exam, certificates

**Learning Paths & Certificates**
Four professional paths — Technician, Engineer, Supervisor, Designer. Complete the required labs, pass the final exam ($\geq 80\%$) and print a certificate with a verification code that any employer can check offline in this program.
NFPA 72 §10.5 Personnel qualificationsNFPA 25 §4.4NICET Level I-IV (reference)

Learning pathsFinal examCertificates & verificationInstructor dashboard

Trainee name (printed on the certificate)Group / company

Save

**Fire Alarm Technician**
Install, test and maintain addressable systems.

0%

Panel lab0% / 60%

Installation0% / 60%

Commissioning0% / 60%

Predictive maintenance0% / 60%

Final examComplete the labs first

**Smart Systems Engineer**
Program, integrate and commission complete smart fire systems.

0%

Panel lab0% / 80%

Cause & effect0% / 80%

BMS integration0% / 60%

Installation0% / 80%

Commissioning0% / 70%

Engineering tools0% / 60%

Final examComplete the labs first

**Fire Safety Supervisor**
Operate the building in emergencies and manage ITM & impairments.

0%

Cause & effect0% / 70%

BMS integration0% / 80%

**Fire Protection Designer**
Design layouts, hydraulics, power and logic to code.

0%

Cause & effect0% / 90%

Installation0% / 90%

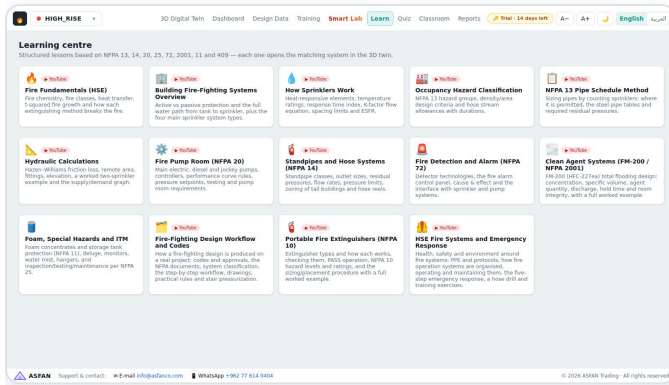
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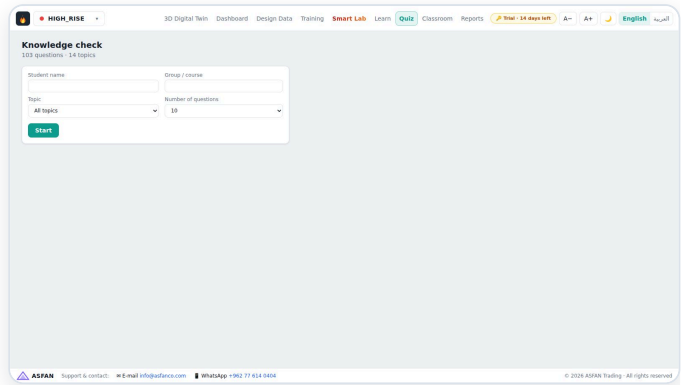
Module 10 – learning paths, final exam and certificates

Fire Protection Digital Twin · User Manual

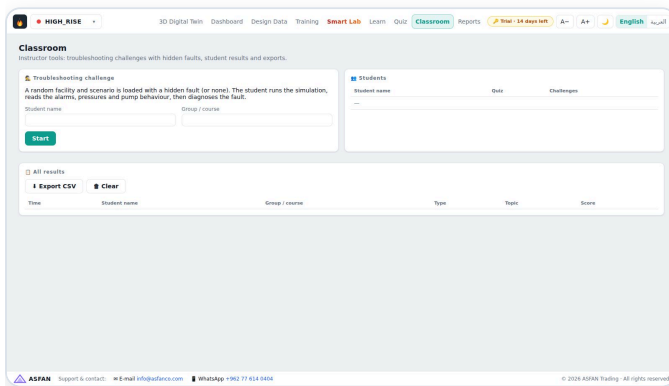
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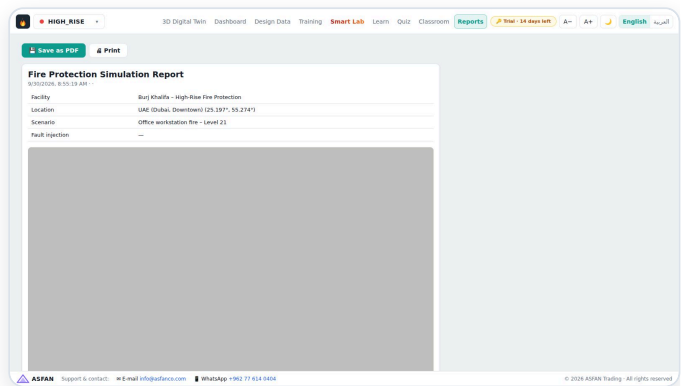
Learn – bilingual lessons with source videos



Quiz – questions with explanations



Classroom – challenges and results



Reports – printable simulation report

- **Learn:** illustrated lessons (fire basics, sprinklers, hazards, hydraulics, pumps, standpipes, detection & alarm, clean agents, foam, ITM) with links to the source videos.
- **Quiz:** choose a topic and number of questions; each answer shows an explanation.
- **Classroom:** enter the student name and group, start a **troubleshooting challenge** (hidden fault), and see results. A supervisor license can export and clear all results (CSV).
- **Reports:** **Save as PDF** or **Print** a full simulation report with screenshot, calculations and event log.

12 Troubleshooting

Problem	Solution
"Windows protected your PC" when installing	Click More info → Run anyway . The program is safe; the message appears because it is not signed with a paid certificate.
3D view is black, slow or flickering	Switch Quality to Low ; update the graphics driver; on laptops choose the dedicated GPU for the program in Windows Graphics settings.
"License is bound to another computer"	Send your current Machine ID to ASFAN to receive a key for this computer.
"System clock rollback detected"	Set the correct date and time in Windows and restart the program.
A tab or module shows 	It is not included in your package. Contact ASFAN to upgrade.
Text too small / large	Use A- / A+ in the top bar.
Results disappeared	Results are stored per Windows user; log in with the same Windows account.

13 Support & contact

ASFAN Trading

E-mail: **info@asfanco.com**

WhatsApp: **+962 77 614 0404**

When you contact support, please send: your name / organisation, your Machine ID (license window), the program version (Help → About) and a screenshot of the problem.